Exercise set 2.3

Q3:- Verify that $\det(kA) = k^n \det(A)$

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix}; \quad k = -2$$

$$\det(A) = \begin{vmatrix} 2 & -1 & 3 & 2 & -1 \\ 3 & 2 & 1 & 3 & 2 \\ 1 & 4 & 5 & 1 & 4 \end{vmatrix}$$

$$= (20 - 1 + 36) - (6 + 8 - 15)$$

$$= 55 + 1 = 56 \quad - (1)$$

$$\text{Now } kA = \begin{bmatrix} -4 & 2 & -6 \\ -6 & -4 & -2 \\ -2 & -8 & -10 \end{bmatrix}$$

So that

$$\det(kA) = \begin{vmatrix} -4 & 2 & -6 & -4 & 2 \\ -6 & -4 & -2 & -6 & -4 \\ -2 & -8 & -10 & -2 & -8 \end{vmatrix}$$

$$= (-160 + 8 - 288) - (-48 - 64 + 120)$$

$$= -440 - 8 = -448 \quad - (2)$$

From (1) and (2), it follows that

$$\det(kA) = -448$$

$$= -8(56)$$

$$= (-2)^3(56)$$

$$= (-2)^3 \det(A)$$

Q6: Vesify That $\det(AB) = \det(BA)$ and determine whether the equality $\det(A+B) = \det(A) + \det(B)$ holds.

$$A = \begin{bmatrix} -1 & 8 & 2 \\ 1 & 0 & -1 \\ -2 & 2 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -1 & -4 \\ 1 & 1 & 3 \\ 0 & 3 & -1 \end{bmatrix}$$

$$A+B = \begin{bmatrix} -1 & 8 & 2 \\ 1 & 0 & -1 \\ 2 & 2 & 2 \end{bmatrix} + \begin{bmatrix} 2 & -1 & -4 \\ 1 & 1 & 3 \\ 0 & 3 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1+2 & 8-1 & 2-4 \\ 1+1 & 0+1 & -1+3 \\ -2+0 & 2+3 & 2-1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 7 & -2 \\ 2 & 1 & 2 \\ -2 & 5 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & 8 & 2 \\ 1 & 0 & -1 \\ -2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & -4 \\ 1 & 1 & 3 \\ 0 & 3 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -2+8+0 & 1+8+6 & 4+24-2 \\ 2+0+0 & -1+0+-3 & -4+0+1 \\ -4+2+0 & 2+2+6 & 8+6-2 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 15 & 26 \\ 2 & -4 & -3 \\ -2 & 10 & 12 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & -1 & -4 \\ 1 & 1 & 3 \\ 0 & 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 8 & 2 \\ 1 & 0 & -1 \\ -2 & 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -2-1+8 & 16+0-8 & 4+1-8 \\ -1+1-6 & 8+0+6 & \cancel{2+4+6} \\ 0+3+2 & 0+0-2 & 2-1+6 \\ & & 0-3-2 \end{bmatrix}$$

$$\text{or } BA = \begin{bmatrix} 5 & 8 & -3 \\ -6 & 14 & 7 \\ 5 & -2 & -5 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} -1 & 8 & 2 \\ 1 & 0 & -1 \\ -2 & 2 & 2 \end{vmatrix}$$

$$= (0+16+4) - (0+2+16)$$

$$= 20 - 18 = 2$$

$$\det(B) = \begin{vmatrix} 2 & -1 & -4 \\ 1 & 1 & 3 \\ 0 & 3 & -1 \end{vmatrix}$$

$$= (-2+0-12) - (0+18+1)$$

$$= -14 - 19 = -33$$

$$\det(A+B) = \begin{vmatrix} 1 & 7 & -2 \\ 2 & 1 & 2 \\ -2 & 5 & 1 \end{vmatrix}$$

$$= (1-28-20) - (4+10+14)$$

$$= -47 - 28 = -75$$

$$\det(AB) = \begin{vmatrix} 6 & 15 & 26 \\ 2 & -4 & -3 \\ -2 & 10 & 12 \end{vmatrix}$$

$$= (-288+90+520) - (208-180+360)$$

$$= 322 - 388$$

$$= -66$$

$$\det(BA) = \begin{vmatrix} 5 & 8 & -3 \\ -6 & 14 & 7 \\ 5 & -2 & -5 \end{vmatrix} \begin{matrix} 5 & 8 \\ -6 & 14 \\ 5 & -2 \end{matrix}$$

$$= (-350 + 280 - 36) - (-210 - 70 + 240)$$

$$= -106 + 40$$

$$= -66$$

From above it can be seen that

$$\det(A+B) = -75 \neq 2 - 33 = \det(A) + \det(B)$$

$$\det(AB) = -66 = \det(A)\det(B)$$

$$\det(BA) = -66 = \det(B)\det(A)$$

Q11:- Use determinants to decide whether the given matrix is invertible.

$$A = \begin{pmatrix} 4 & 2 & 8 \\ -2 & 1 & -4 \\ 3 & 1 & 6 \end{pmatrix}$$

$$\det(A) = \begin{vmatrix} 4 & 2 & 8 \\ -2 & 1 & -4 \\ 3 & 1 & 6 \end{vmatrix} \begin{matrix} 4 & 2 \\ -2 & 1 \\ 3 & 1 \end{matrix}$$

$$= (24 - 24 - 16) - (24 - 16 - 24)$$

$$= -16 + 16$$

$$= 0$$

A is not invertible as $\det(A) = 0$.

Q18:- Find the values of k for which the matrix A is invertible.

$$A = \begin{pmatrix} 1 & 2 & 0 \\ k & 1 & k \\ 0 & 2 & 1 \end{pmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & 2 & 0 \\ k & 1 & k \\ 0 & 2 & 1 \end{vmatrix} \begin{matrix} 1 & 2 \\ k & 1 \\ 0 & 2 \end{matrix}$$

$$= (1 + 0 + 0) - (0 + 2k + 2k)$$

$$= 1 - 4k$$

For A to be invertible it requires that $\det(A) \neq 0$

i.e. $1 - 4k \neq 0$ or $k \neq 1/4$

Q19: Decide whether the matrix is invertible, and if so, use the adjoint method to find its inverse.

$$A = \begin{bmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) \text{ ; if } \det(A) \neq 0$$

$$\operatorname{adj}(A) = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T$$

$C_{11} = \begin{vmatrix} -1 & 0 \\ 4 & 3 \end{vmatrix} = -3 - 0 = -3$ $C_{12} = -\begin{vmatrix} -1 & 0 \\ 2 & 3 \end{vmatrix} = -(-3-0) = 3$

$C_{13} = \begin{vmatrix} -1 & -1 \\ 2 & 4 \end{vmatrix} = -4 + 2 = -2$ $C_{21} = -\begin{vmatrix} 5 & 5 \\ 4 & 3 \end{vmatrix} = -(15-20) = 5$

$C_{22} = \begin{vmatrix} 2 & 5 \\ 2 & 3 \end{vmatrix} = 6 - 10 = -4$ $C_{23} = -\begin{vmatrix} 2 & 5 \\ 2 & 4 \end{vmatrix} = -(8-10) = 2$

$C_{31} = \begin{vmatrix} 5 & 5 \\ -1 & 0 \end{vmatrix} = 0 + 5 = 5$ $C_{32} = -\begin{vmatrix} 2 & 5 \\ -1 & 0 \end{vmatrix} = -(0+5) = -5$

$C_{33} = \begin{vmatrix} 2 & 5 \\ -1 & -1 \end{vmatrix} = -2 + 5 = 3$

To find determinant, use expansion along the 3rd Column

$$\det(A) = 5C_{13} + 0C_{23} + 3C_{33}$$

$$= 5(-2) + 0 + 3(3) = -10 + 9 = -1$$

So inverse exists and is given as $\neq 0$

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} -3 & 3 & -2 \\ 5 & -4 & 2 \\ 5 & -5 & 3 \end{bmatrix}^T = \begin{bmatrix} 3 & -5 & -5 \\ -3 & 4 & 5 \\ 2 & -2 & -3 \end{bmatrix}$$

CamScanner

Q.t:- Solve by Cramers rule, where it applies.

$$x - 4y + z = 6$$
$$4x - y + 2z = -1$$
$$2x + 2y - 3z = -20$$

Cramers rule: if $D \neq 0$

$$x = D_1/D;\ y = D_2/D;\ z = D_3/D$$

$$D = \begin{vmatrix} 1 & -4 & 1 \\ 4 & -1 & 2 \\ 2 & 2 & -3 \end{vmatrix} \begin{matrix} 1 & -4 \\ 4 & -1 \\ 2 & 2 \end{matrix}$$

$$= (3 - 16 + 8) - (-2 + 4 + 48) = -5 - 50 = -55$$

$$D_1 = \begin{vmatrix} 6 & -4 & 1 \\ -1 & -1 & 2 \\ -20 & 2 & -3 \end{vmatrix} \begin{matrix} 6 & -4 \\ -1 & -1 \\ -20 & 2 \end{matrix} = (18 + 160 - 2) - (20 + 24 + 2)$$

$$= 176 - 32 = 144$$

$$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 4 & -1 & 2 \\ 2 & -20 & -3 \end{vmatrix} \begin{matrix} 1 & 6 \\ 4 & -1 \\ 2 & -20 \end{matrix} = (3 + 24 - 80) - (-2 - 40 - 72)$$

$$= -53 + 114 = 61$$

$$D_3 = \begin{vmatrix} 1 & -4 & 6 \\ 4 & -1 & -1 \\ 2 & 2 & -20 \end{vmatrix} \begin{matrix} 1 & -4 \\ 4 & -1 \\ 2 & 2 \end{matrix} = (120 + 8 + 48) - (-12 - 2 + 320)$$

$$= 76 - 306 = -230$$

$$\therefore x = -\frac{144}{55};\ y = -\frac{61}{55}$$

$$z = \frac{230}{55} = \frac{46}{11}$$

Q30:- Show that the matrix

$$A=\begin{bmatrix}\cos\theta & \sin\theta & 0\\ -\sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$$ is invertible

for all values of θ; then find A^{-1} using Theorem (2.3.6)

Theorem 2.3.4 $\left[A^{-1}=\frac{1}{\det(A)}\,adj(A)\right]$

$$C_{11}=\begin{vmatrix}\cos\theta & 0\\ 0 & 1\end{vmatrix}=\cos\theta;\quad C_{12}=-\begin{vmatrix}-\sin\theta & 0\\ 0 & 1\end{vmatrix}=\sin\theta$$

$$C_{13}=\begin{vmatrix}-\sin\theta & \cos\theta\\ 0 & 0\end{vmatrix}=0;\quad C_{21}=\begin{vmatrix}\sin\theta & 0\\ 0 & 1\end{vmatrix}=-\sin\theta$$

$$C_{22}=\begin{vmatrix}\cos\theta & 0\\ 0 & 1\end{vmatrix}=\cos\theta;\quad C_{23}=-\begin{vmatrix}\cos\theta & \sin\theta\\ 0 & 0\end{vmatrix}=0$$

$$C_{31}=\begin{vmatrix}\sin\theta & 0\\ \cos\theta & 0\end{vmatrix}=0;\quad C_{32}=\begin{vmatrix}\cos\theta & 0\\ -\sin\theta & 0\end{vmatrix}=0$$

$$C_{33}=\begin{vmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{vmatrix}=\cos^2\theta+\sin^2\theta=1$$

To find determinant we use expansion along 3rd row

$$\det(A)=0C_{31}+0C_{32}+1C_{33}=1\neq 0$$

so $$A^{-1}=\frac{1}{\det(A)}\begin{bmatrix}C_{11} & C_{12} & C_{13}\\ C_{21} & C_{22} & C_{23}\\ C_{31} & C_{32} & C_{33}\end{bmatrix}^T$$

$$=\begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$$

Q33. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

Assuming that $\det(A) = -7$, find

a. $\det(3A)$ b. $\det(A^{-1})$ c. $\det(2A^{-1})$

d. $\det((2A)^{-1})$ e. $\begin{vmatrix} a & g & d \\ b & h & e \\ c & i & f \end{vmatrix}$

a. $\det(3A) = 3^3 \det(A) = 27(-7) = -189$

b. $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{-7} = -\frac{1}{7}$

c. $\det(2A^{-1}) = 2^3 \det(A^{-1}) = 2^3 / \det(A)$

$= 8/-7 = -8/7$

d. $\det((2A)^{-1}) = \frac{1}{\det(2A)} = \frac{1}{2^3 \det(A)}$

$= \frac{1}{8(-7)} = -\frac{1}{56}$

e. $\begin{vmatrix} a & g & d \\ b & h & e \\ c & i & f \end{vmatrix} = -\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix}$ C_{23}

$= -\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ transpose

$= -(-7) = 7$

Q34- In each part, find the determinant given that A is 4×4 matrix for which $\det(A) = -2$.

a. $\det(-A)$ b. $\det(A^{-1})$ c. $\det(2A^T)$

d. $\det(A^3)$

a. $\det(-A) = (-1)^4 \det(A)$

$= -2$

b. $\det(A^{-1}) = \frac{1}{\det(A)} = -\frac{1}{2}$

c. $\det(2A^T) = 2^4 \det(A^T)$

$= 2^4 \det(A)$

$= 16(-2)$

$= -32$

d. $\det(A^3) = \det(AAA)$

$= \det(A)\det(A)\det(A)$

$= (\det(A))^3$

$= (-2)^3 = -8.$